ECON 6018 TUTORIAL 8

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- Date: 27/03/26

Last Chapter

Unconstrained Optimization

- Multivariate Functions

Same Steps:

1. FOC

2. SOC

Concave (Max)

$f''_{xx} < 0$

$f''_{yy} < 0$

$f''_{xx} f''_{yy} > (f''_{xy})^2$

Convex (Min)

$f''_{xx} > 0$

$f''_{yy} > 0$

$f''_{xx} f''_{yy} > (f''_{xy})^2$

Example:

- Firm's Profit Function

$$\pi(K, L) = 16K^{\frac{1}{4}}L^{\frac{1}{2}} - 2K - L$$

where K is capital and L is labor

Solution

$$\pi(K,L) = 16K^{\frac{1}{4}}L^{\frac{1}{2}} - 2K - L$$

1. FOC:

$$\pi'_K = \frac{1}{4}\cdot 16K^{-\frac{3}{4}}L^{\frac{1}{2}} - 2 = 0$$

$$\pi'_L = \frac{1}{2}\cdot 16K^{\frac{1}{4}}L^{-\frac{1}{2}} - 1 = 0$$

$$\frac{K^{-\frac{3}{4}}L^{\frac{1}{2}}}{2K^{\frac{1}{4}}L^{-\frac{1}{2}}} = 2$$

$$\frac{L}{K} = 4 \qquad L = 4K$$

$$4K^{-\frac{3}{4}}(4K)^{\frac{1}{2}} - 2 = 0$$

$$4K^{-\frac{3}{4}}(2K^{\frac{1}{2}}) = 2$$

$$8K^{-\frac{1}{4}} = 2 \qquad K = \left(\frac{1}{4}\right)^{-4} = 256$$

$L = 4(256) = 1024$

2. SOC

$\pi''_{KK} = 4 \cdot -\frac{3}{4} K^{-\frac{7}{4}} L^{\frac{1}{2}} < 0 \quad = -\frac{3}{512}$

$\pi''_{LL} = 8 \cdot -\frac{1}{2} K^{\frac{1}{4}} L^{-\frac{3}{2}} < 0 \quad = -\frac{1}{2048}$

$\pi''_{KL} = \frac{1}{8} \cdot 16 \cdot K^{-\frac{3}{4}} L^{-\frac{1}{2}} > 0 \quad = \frac{1}{1024}$

$\pi''_{KK} \pi''_{LL} > (\pi''_{KL})^2$

$\therefore$ max

Practice:

$$\pi(K, L) = 18K^{1/4}L^{1/2} - K - 2L$$

Sol$^{\underline{n}}$:

FOC:

- $18 \cdot \frac{1}{4} K^{-\frac{3}{4}} L^{\frac{1}{2}} - 1 = 0$
- $18 \cdot \frac{1}{2} K^{\frac{1}{4}} L^{-\frac{1}{2}} - 2 = 0$

Solve $\dfrac{L}{2K} = \dfrac{1}{2}$

$$L = K$$

$$4.5K^{-\frac{3}{4}}(K)^{\frac{1}{2}} = 1$$

$$K^{-\frac{1}{4}} = \frac{1}{4.5}$$

$$K = \frac{6561}{16}$$

$$L = \frac{6561}{16}$$

SOC

$\pi''_{KK} = (4S) = -\frac{3}{4}K^{-\frac{7}{4}}L^{\frac{1}{2}} < 0$

$\pi''_{LL} = 9 \cdot -\frac{1}{2}K^{\frac{1}{4}}L^{-\frac{3}{2}} < 0$

$\pi''_{KL} = 9 \cdot \frac{1}{4}K^{-\frac{3}{4}}L^{-\frac{1}{2}}$

$\therefore$ max

Application

1. Utility Function

A consumer's utility function is

$$U(x, y) = 12x^{1/2} + 8y^{1/2} - x - 2y$$

where x and y are quantities of two goods.

Question

1. Find the first-order conditions for maximizing $U(x, y)$.
2. Solve for the stationary point (x^*, y^*).
3. Use the second-order test to determine whether the stationary point is a **maximum**, **minimum**, or **neither**.

$U(x, y) = 12x^{1/2} + 8y^{1/2} - x - 2y$

FOC:

$U'_x = 6x^{-\frac{1}{2}} - 1 = 0$

$U'_y = 4y^{-\frac{1}{2}} - 2 = 0$

Solve

$x = \left(\frac{1}{6}\right)^{-2}$ $\quad x = 36$

$y = \left(\frac{1}{2}\right)^{-2}$ $\quad y = 4$

SOC

$U'_x{'}_x = -3x^{-\frac{3}{2}} = -\frac{1}{72} < 0$ $\qquad U'_x{'}_x U'_y{'}_y = \frac{1}{288} > 0$

$U'_y{'}_y = -2x^{-\frac{3}{2}} = -\frac{1}{4}$ $\qquad \therefore$ max

$U'_x{'}_y = 0$

2. Profit Function

A firm produces two goods, x and y. The cost function is

$$C(x, y) = \frac{1}{2}x^2 + xy + y^2 + 2x + y + 10$$

The firm sells the two goods at constant prices $p_x = 20$ and $p_y = 24$.

Question

1. Write down the firm's profit function $\pi(x, y)$.
2. Find the first-order conditions.
3. Solve for the profit-maximizing values x^* and y^*.
4. Find $\dfrac{\partial x^*}{\partial p_x}$ and $\dfrac{\partial y^*}{\partial p_x}$, and interpret the signs.

$$\pi(x,y) = 20x + 24y - \frac{1}{2}x^2 - xy - y^2 - 2x - y - 10$$

FOC

$\pi'_x = 20 - x - y - 2 = 0$ $\quad 18 - x - y = 0 \quad y = 18 - x$

$\pi'_y = 24 - x - 2y - 1 = 0$ $\quad x + 2y = 23$

$x + 2(18 - x) = 23$

$x + 36 - 2x = 23$

$-x = 23 - 36$

$x = 13 \quad y = 5$

$$\pi(x,y) = p_x x + p_y y - \frac{1}{2}x^2 - xy - y^2 - 2xy - 10$$

FOC:

$\pi'_x = p_x - x - y - 2 = 0$

$\pi'_y = p_y - x - 2y - 1 = 0$

$y = p_x - 2 - x$

$p_y - 1 = x + 2y$

$p_y - 1 = x + 2(p_x - 2 - x)$

$p_y - 1 = x + 2p_x - 4 - 2x$

$p_y - 2p_x - 1 + 4 = -x$

$2p_x - p_y - 3 = x$

- Solving for $\frac{\partial x^*}{\partial p_x}$, sub $p_y = 24$

$x^* = 2p_x - 27$

$$\frac{\partial x^*}{\partial p_x} = 2 > 0$$

- Solving for $\frac{\partial y^*}{\partial px}$

$$y^* = px - 2 - (2px - 27)$$

$$= -px + 25$$

$$\frac{\partial y^*}{\partial px} = -1 < 0$$

3. Profit Function

A multiproduct firm sells two products. The inverse demand functions are

$$p_1 = 30 - 2Q_1 \qquad \text{and} \qquad p_2 = 24 - Q_2$$

The cost function is

$$C(Q_1, Q_2) = 4(Q_1 + Q_2)$$

Question

1. Write the firm's total profit function $\pi(Q_1, Q_2)$.
2. Find the profit-maximizing outputs Q_1^* and Q_2^*.
3. Find the corresponding prices p_1^* and p_2^*.
4. Suppose the intercept of product 1's demand rises from 30 to $30 + t$. Find

$$\frac{\partial Q_1^*}{\partial t}, \quad \frac{\partial p_1^*}{\partial t}, \quad \frac{\partial Q_2^*}{\partial t}, \quad \frac{\partial p_2^*}{\partial t}$$

and interpret your answers.

$$\pi(Q_1, Q_2) = 30Q_1 - 2Q_1^2 + 24Q_2 - Q_2^2 - 4(Q_1 + Q_2)$$

FOC:

wrt Q_1: $30 - 4Q_1 - 4 = 0$ $\quad -4Q_1 = -26 \quad Q_1 = \frac{13}{2}$

wrt Q_2: $24 - 2Q_2 - 4 = 0$ $\quad -2Q_2 = -20 \quad Q_2 = 10$

$p_1^* = 30 - 2\left(\frac{13}{2}\right) = 17$

$p_2^* = 24 - 10 = 14$

$$\pi(Q_1, Q_2) = (30+t)Q_1 - 2Q_1^2 + 24Q_2 - Q_2^2 - 4(Q_1 + Q_2)$$

FOC:

wrt Q_1: $(30+t) - 4Q_1 - 4 = 0$ $\quad 26 + t = 4Q_1 \quad Q_1^* = \frac{26+t}{4}$

wrt Q_2: $24 - 2Q_2 - 4 = 0$ $\quad Q_2^* = 10$

$$\frac{\partial Q_1^*}{\partial t} = \frac{1}{4} > 0 \qquad \frac{\partial Q_2^*}{\partial t} = 0$$

$$p_1^* = 30 - 2\left(\frac{26+t}{4}\right)$$

$$\frac{\partial p_1^*}{\partial t} = -\frac{1}{2} < 0$$

$$\frac{\partial p_2^*}{\partial t} = 0$$ p_2^* doesn't contain t